

Safety assessment of the substance 2,2'-oxydiethylamine for use in plastic food contact materials

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The declarations of interest of all scientific experts active in EFSA's work are available at <https://open.efsa.europa.eu/experts>

Abstract

The EFSA Panel on Food Contact Materials assessed the safety of 2,2'-oxydiethylamine, which is intended to be used at up to 14% w/w as a monomer along with adipic acid and caprolactam to make polyamide thin films intended for single use, in contact with all types of food under all conditions of time and temperature. Specific migration of 2,2'-oxydiethylamine was tested from a polyamide film in water and was below the limit of quantification (LOQ) of 0.015 mg/kg. Migration of impurities has been estimated pro-rata. Migrating oligomers were identified and semi-quantified in 10% ethanol, 3% acetic acid and olive oil. Considering the measured migration and the virtual presence at the LOQ of the oligomers below 1000 Da containing the substance, the estimated migration of these oligomers would be 1.042 mg/kg. Genotoxicity studies were performed on the substance and on 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione and 1-oxa-4,11,18,25-tetraazacycloheptacosane-5,10,17,24-tetraone. (Q)SAR analyses were provided on the oligomers of higher molecular masses. Based on these data, the Panel excluded genotoxicity concerns for the substance and its oligomers. From a 90-day toxicity study on the 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione, the Panel identified a NOAEL of 1040 mg/kg bw per day. Based on their physico-chemical properties and experimental data, the Panel considered the potential for accumulation in humans of the oligomers containing the substance of no concern. The Panel concluded that the substance is not of safety concern for the consumer if it is used as a comonomer with 99.6% minimum purity at up to 14% w/w to manufacture polyamide films (maximum thickness: 25 µm) and intended to be used in contact with all types of foods, except infant formula and human milk, at all time and temperature conditions. The migrations of the substance and of the oligomers below 1000 Da containing the substance should not exceed 0.05 and 5 mg/kg food, respectively.

KEYWORDS

2,2'-oxydiethylamine, caprolactam, comonomer, FCM substance No 1094, film, food contact materials, oligomers, polyamide, safety assessment

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1 | INTRODUCTION

1.1 | Background and Terms of Reference

Before a substance is authorised to be used in food contact materials (FCM) and is included in a positive list, the European Food Safety Authority (EFSA)'s opinion on its safety is required. This procedure has been established in Articles 8, 9 and 10 of Regulation (EC) No 1935/2004¹ of the European Parliament and of the Council of 27 October 2004 on materials and articles intended to come into contact with food.

According to this procedure, the industry submits applications to the competent authorities of Member States, which transmit the applications to the (EFSA) for evaluation.

In this case, EFSA received an application from the German competent authority (Bundesamt für Verbraucherschutz und Lebensmittelsicherheit, BVL), requesting the safety evaluation of the substance 2,2'-oxydiethylamine, FCM substance No 1094, with the CAS number 2752-17-2. The request has been registered in EFSA's register of received questions under the number EFSA-Q-2024-00102. The dossier was submitted by BASF SE.

According to Regulation (EC) No 1935/2004 of the European Parliament and of the Council on materials and articles intended to come into contact with food, EFSA is asked to carry out an assessment of the risks related to the intended use of the substance and to deliver a scientific opinion.

2 | DATA AND METHODOLOGIES

2.1 | Data

The applicant has submitted a confidential and a non-confidential version of a dossier following the 'EFSA Note for Guidance' for the preparation of an application for the Safety Assessment of a Substance to be used in Plastic Food Contact Materials (EFSA CEF Panel, 2008) and the 'Administrative guidance for the preparation of applications on substances to be used in plastic food contact materials' (EFSA, 2021).

In accordance with Art. 38 of the Commission Regulation (EC) No 178/2002² and taking into account the protection of confidential information and of personal data in accordance with Articles 39–39e of the same Regulation and of the Decision of the EFSA's Executive Director laying down practical arrangements concerning transparency and confidentiality,³ the non-confidential version of the dossier is published on Open.EFSA.⁴

According to Art. 32c(2) of Regulation (EC) No 178/2002 and to the Decision of EFSA's Executive Director laying down the practical arrangements on pre-submission phase and public consultations³, EFSA carried out a public consultation on the non-confidential version of the application from 23 July to 13 August 2024, for which no comments were received.

Data submitted and used for the evaluation are:

Non-toxicological data and information

- Chemical identity
- Description of manufacturing process of substance
- Physical and chemical properties
- Intended applications
- Migration of the substance
- Residual content of the substance in the FCM
- Identification, quantification and migration of oligomers and reaction products

Toxicological data

2,2'-oxydiethylamine

- in vitro bacterial reverse mutation assay
- in vitro mammalian cell micronucleus test

¹Regulation (EC) No 1935/2004 of the European parliament and of the council of 27 October 2004 on materials and articles intended to come into contact with food and repealing Directives 80/590/EEC and 89/109/EEC. OJ L 338, 13.11.2004, p. 4–17.

²Regulation (EC) No 178/2002 of the European Parliament and of the Council of 28 January 2002 laying down the general principles and requirements of food law, establishing the European Food Safety Authority and laying down procedures in matters of food safety. OJ L 31, 1.2.2002, p. 1–48.

³Decision available at <https://www.efsa.europa.eu/en/corporate-pubs/transparency-regulation-practical-arrangements>.

⁴The non-confidential version of the dossier, following EFSA's assessment of the applicant's confidentiality requests, is published on Open.EFSA and is available at the following link: <https://open.efsa.europa.eu/dossier/FCM-2023-14391>.

Oligomer 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione ($x = 1$; $y = 1$), constituted by adipic acid and 2,2'-oxydiethylamine (together as $x = 1$) and caprolactam ($y = 1$)

- in vitro bacterial reverse mutation assay
- in vitro mammalian cell micronucleus test
- in silico data
- 14-day oral toxicity study in rats
- 28-day oral toxicity study in rats
- Plasma level determination after oral repeated 28-day treatment in rats
- 90-day oral toxicity study in rats

Oligomer 1-oxa-4,11,18,25-tetraazacycloheptacosane-5,10,17,24-tetraone ($x = 1$; $y = 2$)

- in vitro bacterial reverse mutation assay
- in vitro mammalian cell micronucleus test
- in silico data

other oligomers ($x = 1$; $y = 3-6$)

- in silico data

2.2 | Methodologies

The assessment was conducted in line with the principles laid down in Regulation (EC) No 1935/2004 on materials and articles intended to come into contact with food. This Regulation underlines that applicants may consult the Guidelines of the Scientific Committee on Food (SCF) for the presentation of an application for safety assessment of a substance to be used in FCM prior to its authorisation (European Commission, 2001), including the corresponding data requirements. The dossier that the applicant submitted for evaluation was in line with the SCF guidelines (European Commission, 2001).

The methodology is based on the characterisation of the substance that is the subject of the request for safety assessment prior to authorisation, its impurities and reaction and degradation products, the evaluation of the exposure to those substances through migration and the definition of minimum sets of toxicity data required for safety assessment.

To establish the safety from ingestion of migrating substances, the toxicological data indicating the potential hazard and the likely human exposure data need to be combined. Exposure is estimated from studies on migration into food or food simulants and considering that a person may consume daily up to 1 kg of food in contact with the relevant FCM.

As a general rule, the greater the exposure through migration, the more toxicological data is required for the safety assessment of a substance. Currently, there are three tiers with different thresholds triggering the need for more toxicological information as follows:

- In case of high migration (i.e. 5–60 mg/kg food), an extensive data set is needed.
- In case of migration between 0.05 and 5 mg/kg food, a reduced data set may suffice.
- In case of low migration (i.e. <0.05 mg/kg food), only a limited data set is needed.

More detailed information on the required data is available in the SCF guidelines (European Commission, 2001).

The assessment was conducted in line with the principles described in the EFSA Guidance on transparency in the scientific aspects of risk assessment (EFSA, 2009) and considering the relevant guidance from the EFSA Scientific Committee.

3 | ASSESSMENT

According to the applicant, the substance 2,2'-oxydiethylamine is intended to be used at up to 14% w/w as a monomer along with adipic acid and caprolactam to manufacture polyamides. The polymer is intended to be used as single-use films of up to 25 μm thickness in contact with all types of food under all conditions of time and temperature. Reported applications are for packaging meat, ripening fermented dry sausage, fruits and vegetables as well as curing, e.g. meat by heating (6 h at 100°C) and subsequent long-term storage. The uses requested by the applicant do not specify contact with infant formula and human milk.

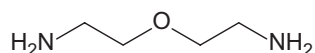
The substance has not been evaluated in the past by the SCF or EFSA. Adipic acid and caprolactam are both authorised according to Reg. (EU) 10/2011 as FCM substance No 303 without restrictions and as FCM substance No 212 with a specific migration limit (SML) of 15 mg/kg food, respectively.

3.1 | Non-toxicological data

3.1.1 | Identity of the substance⁵

Chemical formula: C₄H₁₂N₂O; molecular mass: 104.15 Da

Chemical structure:



The substance is manufactured by reacting diethylene glycol with ammonia at elevated temperature and pressure in the presence of a catalyst. The product is purified through four distillation steps.

The specified purity is ≥ 99.6% w/w; identified impurities are ethanolamine (0.2% w/w), morpholine (0.01% w/w) and 3,6-dihydro-2H-1,4-oxazine (0.14% w/w).

Considering the impurities identified, ethanolamine is authorised in accordance with Regulation (EU) 10/2011 as FCM substance No 326 (2-aminoethanol) with an SML of 0.05 mg/kg food and the following restrictions: 'Not to be used for articles in contact with fatty foods for which simulant D1 and/or D2 is laid down. For indirect food contact only, behind a PET layer'. Morpholine and 3,6-dihydro-2H-1,4-oxazine are not listed in the Annex I of Reg. (EU) 10/2011.

3.1.2 | Physical and chemical properties⁶

The substance has a melting point ranging from −48 to −43°C, a boiling point of 186°C and a decomposition temperature of 415°C, the last measured by differential scanning calorimetry (DSC). Hence, no relevant thermal degradation of the substance is expected at the provided maximum temperature of manufacture of the intended polyamide plastics (250°C). The substance is miscible with water at room temperature and has a log Po/w at 23°C of −1.42 at pH 6.2 and −1.37 at pH 11.

3.1.3 | Residual content in the plastic, specific migration of the substance and overall migration

A polyamide film (25 µm thickness, density 1.163 g/cm³) made with a nominal starting content of 13% w/w of 2,2'-oxydiethylamine was used as a representative sample for measuring the residual content in the film as well as the specific and the overall migrations.

The residual content of the substance⁷ was below the limit of quantification (LOQ) of 0.4 mg/kg polymer, which corresponds to 0.1 µg/dm² for the film of 25 µm thickness.

The specific migration⁸ of 2,2'-oxydiethylamine from the polyamide film was tested for 6 h at 100°C followed by 10 days at 60°C only in water, because under the applied contact conditions, the substance was unstable in 10% ethanol (food simulant A) and reactive with components of olive oil (food simulant D2). Moreover, the film was physically damaged when tested in 3% acetic acid (food simulant B) for 10 days at 60°C. Although it was possible to test the film in 3% acetic acid for 10 days at 40°C (as foreseen in Annex V, Chapter 2, section 2.1.3 of Reg. (EU) 10/2011), water was chosen, since higher test temperatures could be reached. Considering the high solubility of the substance in water and the possibility for testing at 60°C instead of 40°C, the Panel agreed that contact with water under these test conditions was the worst-case scenario for the migration of 2,2'-oxydiethylamine. The migration was measured by LC-MS and, after correcting for the recovery from the stability test (which was 40% in water), it was below the LOQ of 0.0025 mg/dm², which corresponds to 0.015 mg/kg food when applying the standard surface to volume (S/V) ratio of 6 dm²/kg.

Overall migration tests by total immersion were performed for 4 h at reflux temperature in 3% acetic acid and 10% ethanol and for 4 h at 100°C in olive oil. The results were 6.8 mg/dm² and 6.4 mg/dm² in 3% acetic acid and 10% ethanol, respectively, while not detectable (< 3 mg/dm²) in olive oil.

3.1.4 | Migration of impurities, comonomers and oligomers related to the substance

The migration of impurities was not tested. The impurities are expected to be reactive in the polymerisation process and to become chemically bound to the polymer. Based on the level of impurities in the substance reported in Section 3.1.1, the FCM Panel estimated their potential migration pro-rata to the migration of the substance itself (< LOQ 0.015 mg/kg food).

⁵Technical dossier, section 'Identity of the substance'.

⁶Technical dossier, section 'Physical and chemical properties of substance'.

⁷Technical dossier, section 'Data on residual content of substance in the food contact material'.

⁸Technical dossier, section 'Data on Migration of substance'.

The results of the potential migrations⁹ were below 3×10^{-6} , 6×10^{-5} and 4.2×10^{-5} mg/kg food for morpholine, ethanolamine and 3,6-dihydro-2H-1,4-oxazine, respectively; so, well below the SML of 0.05 mg/kg food for ethanolamine. If the migration of the substance itself is restricted to 0.05 mg/kg food, the migration of the three impurities, if any, would correspondingly be multiplied by 3.3 (being 0.05/0.015), that is 1×10^{-5} , 2×10^{-4} and 1.4×10^{-4} mg/kg food.

The migration of the comonomer caprolactam, measured from the same film as described in Section 3.1.3, was lower than the LOQ (0.0043 mg/dm² or 0.026 mg/kg food) after 10 days at 40°C in 3% acetic acid, while it was measurable in the other simulants, tested for 6 h at 100°C followed by 10 days at 60°C: in 10% ethanol it was 0.0148 mg/dm² or 0.0887 mg/kg food, in olive oil 0.0027 mg/dm² or 0.0165 mg/kg food. These results are well below the SML for caprolactam of 15 mg/kg. The method for the aqueous simulants was based on the UV signal of a LC-UV-MS analysis. For olive oil, LC-MS was used.

The migration of the oligomers was tested from the same film described in Section 3.1.3. The oligomers identified by LC-HRMS were used to characterise the migrated oligomeric fraction. The oligomers below 1000 Da containing 2,2'-oxydiethylamine, identified as 'mixed oligomers', were made with one unit of 2,2'-oxydiethylamine and adipic acid (together as $x = 1$) plus 1 to 6 units of caprolactam ($y = 1-6$). 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione (cyclic amide 1, $x = 1$; $y = 1$) is the smallest cyclic oligomer that contains each of the three monomers. It was used as a reference substance in migration testing (Figure 1).

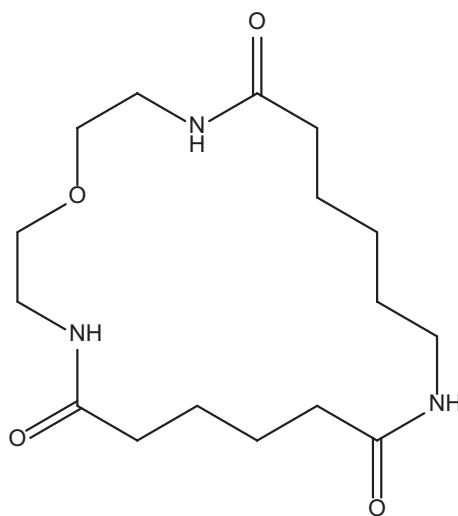


FIGURE 1 Chemical structure of 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione (cyclic amide 1, $x = 1$; $y = 1$), with chemical formula: $C_{16}H_{29}N_3O_4$.

The migration of the cyclic amide 1 was tested for 10 days at 40°C in 3% acetic acid as well as for 6 h at 100°C followed by 10 days at 60°C in 10% ethanol and olive oil (as for caprolactam). It was below the LOD of 0.0063 mg/dm² (0.038 mg/kg food) in both 3% acetic acid and 10% ethanol, measured by UV detection. In olive oil, the migration measured by LC-MS was 0.0006 mg/dm² (or 0.0037 mg/kg food), just above the LOQ.

The migration of mixed oligomers below 1000 Da (including cyclic amide 1) from a film sample identical to the one described in Section 3.1.3 was tested after 10 days at 40°C in 3% acetic acid as well as after 6 h at 100°C followed by 10 days at 60°C in both 10% ethanol and olive oil. In 3% acetic acid and 10% ethanol, the migration was determined using LC-UV, assuming that the UV response was the same for all oligomers and using external calibration with the cyclic amide 1 as the standard analogue. In olive oil, the migration was measured by LC-MS and semi-quantified using as standard a 95% ethanol extract of the polymer that was previously characterised and semi-quantified using cyclic amide 1 and UV detection. The lower bound migrations (i.e. excluding results $\leq$ LOQ and/or $\leq$ LOD) of these mixed oligomers were 0.526, 0.820 and 0.0037 mg/kg food in 3% acetic acid, 10% ethanol and olive oil, respectively. Their upper-bound migration (i.e. including numerically both the LOQ when migration was below LOQ and the measured value that was between LOQ and LOD) in 10% ethanol (considered as worst case) would be 1.042 mg/kg food.

The Panel noted that the applied migration testing conditions did not fully cover those set in Reg. (EU) 10/2011 for high-temperature applications (above 200°C), such as for use in ovens. However, considering that for a film of 25 μ m, the contact conditions (6 h at 100°C followed by 10 days at 60°C) cause an extraction rather than a migration, under oven conditions the migration is not expected to be higher.

⁹To allow for a potential lower reactivity of the three 'monoamine' impurities compared to the diamine substance, the Panel introduced an additional factor 2 in this calculation.

3.2 | Toxicological data¹⁰

In accordance with the 'EFSA Note for Guidance' (EFSA CEF Panel, 2008), to assess the genotoxic potential of the substance (migration below 0.05 mg/kg food), the applicant submitted a bacterial reverse mutation assay (Ames test) and an in vitro mammalian cell micronucleus test conducted on 2,2'-oxydiethylamine.

The upper-bound migration of the mixed oligomers below 1000 Da ($x=1$; $y=1-6$) was 1.042 mg/kg food. Therefore, according to the 'EFSA Note for Guidance' (EFSA CEF Panel, 2008), their potential genotoxicity and general toxicity should be addressed (migration between 0.05 and 5 mg/kg food). Two bacterial reverse mutation assays (Ames test), two in vitro mammalian cell micronucleus tests as well as in silico analyses carried out, respectively, with either cyclic amide 1 ($x=1$; $y=1$) or 1-oxa-4,11,18,25-tetraazacycloheptacosane-5,10,17,24-tetraone ($x=1$; $y=2$) were submitted. The Panel considered that these two oligomers are representative of the migrating oligomers with molecular mass below 1000 Da. The genotoxic potential of the remaining mixed oligomers ($x=1$; $y=3-6$) was assessed in silico. Cyclic amide 1 ($x=1$; $y=1$) was tested in a 90-day study in rats as well as in a 14-day and in a 28-day oral range-finding toxicity studies in rats. Plasma levels of cyclic amide 1 were determined from the latter.

3.2.1 | Genotoxicity

3.2.1.1 | Impurities

The impurity ethanolamine, FCM Substance No 326, would migrate (if any) below its SML of 0.05 mg/kg food and therefore is not addressed further.

As explained in Section 3.1.4, the impurities morpholine and 3,6-dihydro-2H-1,4-oxazine might migrate, if at all, below 0.15 µg/kg food, hence leading to a potential exposure below the threshold of toxicological concern (TTC) of 0.0025 µg/kg body weight per day (EFSA Scientific Committee, 2012a). However, for morpholine, the FCM Panel also calculated the risk from the potential endogenous formation of N-nitrosomorpholine, using the margin of exposure (MoE) approach (EFSA Scientific Committee, 2012b), because the TTC should not be used for N-nitroso-compounds (EFSA Scientific Committee, 2012a). According to the model for endogenous nitrosamines formation described in the EFSA 'Re-evaluation of potassium nitrite (E249) and sodium nitrite (E250)' (EFSA ANS Panel, 2017), the daily endogenous formation of N-nitrosomorpholine from morpholine migrating as an impurity in 2,2'-oxydiethylamine was calculated and resulted in 1.3×10^{-14} mg/kg bw per day. The BMDL₁₀ for N-nitrosomorpholine being 0.7 mg/kg bw per day (SCCS, 2012), the resulting margin of safety was estimated to be $\sim 5 \times 10^{13}$. Based on this result, the Panel concluded that the potential migration (if any) of morpholine does not raise any safety concern related to the potential formation of N-nitrosomorpholine in the stomach.

3.2.1.2 | 2,2'-oxydiethylamine

3.2.1.2.1 | Bacterial reverse mutation assay

A bacterial reverse mutation assay was conducted in *Salmonella Typhimurium* strains TA98, TA100, TA1535, TA1537 and in *Escherichia coli* WP2 *uvrA* to assess the mutagenicity of 2,2'-oxydiethylamine. The study was performed in compliance with good laboratory practice (GLP) principles, following the Organisation for Economic Co-operation and Development (OECD) TG 471 (OECD, 2020). The test material (purity 99.7%) was dissolved in ultrapure water and evaluated in two separate experiments: the first one using the plate incorporation method, and the second one using the preincubation procedure. Both assays were performed in the absence and presence of metabolic activation by phenobarbital/β-naphthoflavone-induced rat liver S9 fraction (S9-mix). The following six concentrations: 33, 100, 333, 1000, 2500 and 5000 µg/plate were tested in both experiments, with and without S9. All concentrations, as well as negative (vehicle) and positive controls, were evaluated using triplicate plates. No precipitation of the test substance was observed. In the first experiment, using the plate incorporation method, a reduction of the background growth of auxotrophic bacteria, indicating a bacteriotoxic effect, was observed at 5000 µg/plate in all strains with and without S9. In the repeated experiment with the preincubation procedure, a reduction of background growth and of the number of revertant colonies was observed at 1000 µg/plate or above, depending on the strain and test conditions. No increase in revertant colonies was observed at any concentration in any bacterial tester strain, with or without metabolic activation. Both vehicle and positive controls were within the respective historical control ranges.

The test item did not induce gene mutations under the conditions of the study. Based on the 'Harmonised approach for reporting reliability and relevance of genotoxicity studies' (EFSA, 2023), the Panel considered the study reliable without restrictions and the results of high relevance.

¹⁰Technical dossier, section 'Toxicological data'.

3.2.1.2.2 | *In vitro mammalian cell micronucleus test*

The substance 2,2'-oxydiethylamine (purity 99.7%) was tested in the *in vitro* micronucleus assay in human peripheral blood lymphocytes using the cytokinesis-block method. The test was performed in compliance with GLP principles and according to the OECD TG 487 (OECD, 2016). Duplicate lymphocyte cultures from healthy donors were treated with the test item, dissolved in culture medium, for 4 h with and without phenobarbital/ β -naphthoflavone-induced rat liver S9, followed by 16 h recovery in the presence of cytochalasin B, or for 20 h without S9 in the presence of cytochalasin B. Based on the results of a pretest on solubility and osmolality, the following three doses were selected for the scoring of micronuclei in 1000 binucleated cells/culture: 188.6, 611.1, 1100 $\mu\text{g/mL}$ (~ 10 mM, maximum recommended concentration). The cytotoxicity of treatments was evaluated determining the cytokinesis-block proliferation index (CBPI) in 500 cells per culture. No precipitation of the test item was observed at the end of the incubation period. No increase in binucleated cells containing micronuclei and no dose-related toxicity were observed in treated cultures compared to vehicle controls under any experimental condition.

The test item did not induce the formation of micronuclei under the conditions of the study. The Panel considered the study reliable without restrictions and the results of high relevance.

3.2.1.3 | *1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione (cyclic amide 1, x = 1; y = 1)*

3.2.1.3.1 | *Bacterial reverse mutation assay*

The same protocol applied for 2,2'-oxydiethylamine was used to evaluate the mutagenicity of 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione. The test material (purity > 95%) was dissolved in ultrapure water. Six concentrations (33, 100, 333, 1000, 2650 and 5300 $\mu\text{g/plate}$) were tested in both experiments, with and without S9. No precipitation of the test substance was observed. No relevant bacteriotoxic effect and no increase in revertant colonies were observed at any dose in any bacterial tester strain, with or without metabolic activation. Both vehicle and positive controls were within the respective historical control ranges.

The test item did not induce gene mutations under the conditions of the study. The Panel considered the study reliable without restrictions and the results of high relevance.

3.2.1.3.2 | *In vitro mammalian cell micronucleus test*

The substance 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione was tested in the *in vitro* micronucleus assay in human peripheral blood lymphocytes. The test was performed following the same protocol applied for 2,2'-oxydiethylamine. The test substance (purity > 95%) was dissolved in culture medium. The following three doses were selected for the scoring of micronuclei: 679, 1222.2, 2200 $\mu\text{g/mL}$ ($\approx$ maximum recommended concentration by weight). No precipitation of the test item was observed at the end of the incubation period. No increase in binucleated cells containing micronuclei and no dose-related toxicity were observed in treated cultures compared to vehicle controls under any experimental condition.

The test item did not induce the formation of micronuclei under the conditions of the study. The Panel considered the study reliable without restrictions and the results of high relevance.

3.2.1.4 | *1-oxa-4,11,18,25-tetraazacycloheptacosane-5,10,17,24-tetraone (x = 1; y = 2)*

3.2.1.4.1 | *Bacterial reverse mutation assay*

The same protocol applied for 2,2'-oxydiethylamine was used to evaluate the mutagenicity of 1-oxa-4,11,18,25-tetraazacycloheptacosane-5,10,17,24-tetraone. The test material (purity > 96%) was suspended in dimethyl sulfoxide and sonicated before testing to achieve a homogeneous fine suspension. Six concentrations (33, 100, 333, 1000, 2500 and 5000 $\mu\text{g/plate}$) were tested in both experiments, with and without S9. No precipitation of the test substance was observed. No bacteriotoxic effect and no increase in revertant colonies were observed at any dose in any bacterial tester strain, with or without metabolic activation. Both vehicle and positive controls were within the respective historical control ranges.

The test item did not induce gene mutations under the conditions of the study. The Panel considered the study reliable without restrictions and the results of high relevance.

3.2.1.4.2 | *In vitro mammalian cell micronucleus test*

The substance 1-oxa-4,11,18,25-tetraazacycloheptacosane-5,10,17,24-tetraone was tested in the *in vitro* micronucleus assay in human peripheral blood lymphocytes. The test was performed following GLP principles and according to the OECD TG 487 (OECD, 2016), with the same protocol as applied for 2,2'-oxydiethylamine. The test substance (purity > 96%) was dissolved in culture medium by sonication. The following three doses were selected for the scoring of micronuclei: 190.5, 1111.1 and 2000 $\mu\text{g/mL}$ (maximum recommended concentration) with short (4 h) treatment; 342.9, 1111.1 and 2000 $\mu\text{g/mL}$ with the continuous (20 h) treatment. No precipitation of the test item was observed at the end of the incubation period.

No increase in binucleated cells containing micronuclei and no dose-related toxicity were observed in treated cultures compared to vehicle controls under any experimental condition.

The test item did not induce the formation of micronuclei under the conditions of the study. The Panel considered the study reliable without restrictions and the results of high relevance.

3.2.1.5 | *Mixed oligomers ($x=1$; $y=3-6$)*

The genotoxic potential of the mixed oligomers below 1000 Da containing the substance was also evaluated using in silico models for the prediction of the activity in the Ames test (DEREK and TIMES) and the induction of structural chromosomal aberrations in mammalian cells (TIMES). For all structures of the mixed oligomers ($x=1$; $y=3-6$), a negative prediction for mutagenicity in the Ames test and the activity in the structural chromosomal aberration test were formulated.

3.2.1.6 | *Conclusions on genotoxicity*

The test substance 2,2'-oxydiethylamine as well as cyclic amide 1 ($x=1$; $y=1$), and 1-oxa-4,11,18,25-tetraazacycloheptacosane-5,10,17,24-tetraone ($x=1$; $y=2$) were evaluated in the recommended set of in vitro genotoxicity studies (Ames and micronucleus tests) with highly relevant negative results. Furthermore, higher molecular mass mixed oligomers ($x=1$; $y=3-6$) showed negative in silico predictions for in vitro mutagenicity (Ames test) and structural chromosomal aberrations. Therefore, based on the available experimental and in silico data and considering the structural similarity of all oligomers, the Panel concluded that the substance and the oligomers containing the substance formed under the reported conditions do not raise a concern for genotoxicity.

3.2.2 | General toxicity

The oligomer 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione (cyclic amide 1, $x=1$; $y=1$) was tested in 14-day, 28-day and 90-day oral repeated toxicity studies. The Panel considered that the oligomer is representative of the migrating oligomers with molecular masses below 1000 Da. The accumulation potential of the oligomer was evaluated in the 28-day repeated-dose oral toxicity study in rats, with a plasma concentration analysis.

3.2.2.1 | *Repeated dose 14-day and 28-day range-finding oral toxicity studies in rats*

A 14-day dietary range-finding oral toxicity study (not following any OECD TG, not in compliance with GLP principles) was performed in rats with 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione (cyclic amide 1, $x=1$; $y=1$). No significant signs of systemic toxicity were detected up to the higher concentration tested. Based on this range-finding study, the concentration to be tested in the 28-day oral in-feed toxicity study were selected as being 0, 1500, 5000 and 15,000 ppm.

1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione (batch n° 21IG6166; purity >95%, certificate of analysis provided) was tested in a 28-day oral subchronic toxicity study in Wistar Crl:WI(Han) rats, according to the OECD TG 407 (OECD, 2008) and in compliance with GLP principles (2023). Four groups of animals (5/sex/group) were treated daily via the diet at concentrations of 0 ppm (control group), 1500 ppm (140 and 141 mg/kg bw per day in males and females, respectively), 5000 ppm (459 and 457 mg/kg bw per day in males and females, respectively) and 15,000 ppm (1401 and 1376 mg/kg bw per day in males and females, respectively).

No test substance-related signs of systemic toxicity were detected up to a concentration of 15,000 ppm. Functional performance tests did not indicate any effect of treatment. No test substance-related alterations in haematology, clinical biochemistry and urinalysis parameters were observed in any test group. No changes in organ weights as well as macroscopic and histopathological findings were observed. Few findings (i.e. minimal increase in size and number of follicular cells and alteration in colloid of thyroid glands) in two males of the high-dose group were considered non-adverse, in the absence of any other alteration.

Based on this range-finding study, the concentration levels to be used in the 90-day oral toxicity study were selected as being identical to those tested in the 28-day toxicity study.

3.2.2.2 | *Repeated dose 90-day oral toxicity study in rats*

1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione (batch no IG6176; Purity: cyclic amide 1 98.4% w/w; adipic acid 0.01% w/w; bis(2-aminoethyl)ether 0.01% w/w; caprolactam 0.01% w/w) was tested in a 90-day subchronic oral toxicity study in Wistar Crl:WI(Han) rats, according to OECD TG 408 (OECD, 2018) and in compliance with GLP principles (2023). Four groups of animals (10/sex/group) were treated daily via the diet at concentrations of 0 ppm (control group), 1500 ppm (95 and 113 mg/kg bw per day in males and females, respectively), 5000 ppm (347 and 388 mg/kg bw per day in males and females, respectively) and 15,000 ppm (1040 and 1209 mg/kg bw per day in males and females, respectively).

There were no unscheduled deaths throughout the study. No changes in food and water consumption, as well as clinical signs and behavioural effects were observed at any dose level during the study. No test substance-related changes in body weights were observed in males and females of any test group. Functional performance tests including home cage

and open field observations and sensorimotor tests/reflexes as well as measurement of motor activity did not indicate any effect of treatment. Ophthalmic examination did not indicate any treatment-related effect.

No test substance-related alterations in haematology, clinical biochemistry and urinalysis parameters were observed in male and female animals of any test group. In males of mid-dose group, relative eosinophil cell counts were increased, but the change was not dose dependent. Therefore, this alteration was regarded as incidental and not treatment related. In males of high-dose group alkaline phosphatase (ALP) activities were decreased ($1.1 \pm 0.2 \mu\text{kat/L}$), but the values were within the historical control range (males, ALP 0.93–1.49 $\mu\text{kat/L}$). Therefore, this change was regarded as incidental and not treatment related. No test substance-related changes in thyroid hormone (i.e., T3, T4 and TSH) levels were observed.

No macroscopic findings were observed at necropsy that might be considered associated with treatment. Due to the absence of any histopathological finding and dose–response relationship, decreases in the relative weight of the spleen (–14.8%), significant only in males of the mid dose group and in the relative weight of the adrenal (–18.4%, only at the low dose) and pituitary (–23.4%, –13.5% at the low and high doses, respectively) glands in females were considered unrelated to treatment.

No histopathological findings were reported. Special attention was also given for the synchrony of the morphology of the oestrous cycle in ovaries, uterus, cervix and vagina, and no findings were reported.

In the absence of treatment-related adverse effects up to the highest concentration tested, under the conditions of the present study, the Panel identified a no-observed adverse effect level (NOAEL) of 1040 mg/kg bw per day (in males), the highest dose tested.

3.2.2.3 | Plasma level determination of 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione (cyclic amide 1) after oral repeated 28-day treatment in Wistar rats

During the repeated dose 28-day oral toxicity study reported in Section 3.2.2.1 (not GLP-compliant for this experimental phase only) (2023), after daily treatment in the diet (0, 500, 1500 and 15,000 ppm) of Wistar rats with cyclic amide 1, blood samples were collected on days 7 and 23. Analytical determination of the test substance by LC–MS measurements (LOD 1.8 ng/mL) was performed in samples from all dose group animals. The test substance was not detected in the plasma from the controls. The plasma levels of the test substance increased with the increasing doses. The measured plasma concentrations of the substance were comparable on days 7 and 23 for both females and males and at each dose levels. This suggests that a plateau is reached after 7 days.

3.3 | Discussion

No migration of the substance 2,2'-oxydiethylamine was detected at the LOQ of 0.015 mg/kg. The sum of the migrating mixed oligomers ($x=1$; $y=1-6$) with molecular masses below 1000 Da and assuming their virtual presence (upper-bound assumption) at the LOD/LOQ was up to 1.042 mg/kg, hence below 5 mg/kg food.

Based on the results of in vitro bacterial reverse mutation assays and in vitro mammalian cell micronucleus tests carried out each on the 2,2'-oxydiethylamine and the two representative mixed oligomers, 1-oxa-4,11,18-triazacycloeicosane-5,10,17-trione (cyclic amide 1, $x=1$; $y=1$) and 1-oxa-4,11,18,25-tetraazacycloheptacosane-5,10,17,24-tetraone ($x=1$; $y=2$), the FCM Panel concluded that they did not raise a concern for genotoxicity. (Q)SAR analyses on the other mixed oligomers ($x=1$; $y=3-6$) indicated the lack of genotoxic potential. Overall, the Panel concluded that the substance and the oligomers containing the substance, formed under the reported conditions, do not raise a concern for genotoxicity.

Potential individual migration of the three reported impurities was estimated to be below 0.06 $\mu\text{g/kg}$ and considered of no safety concern. Ethanolamine (0.2% w/w) was assessed with no concern for genotoxicity (SCF, 2000). The migration of 3,6-dihydro-2H-1,4-oxazine (0.14% w/w) and morpholine (0.01% w/w), if any, would be below 0.15 $\mu\text{g/kg}$ food, leading to a potential exposure below the TTC of 0.0025 $\mu\text{g/kg}$ body weight (bw) per day (EFSA Scientific Committee, 2012a). The potential migration of morpholine does not raise safety concern related to the potential formation of N-nitrosomorpholine in the stomach (estimated MoS of $\sim 5 \times 10^{13}$) (EFSA Scientific Committee, 2012b).

Based on the results of the repeated dose 90-day oral toxicity study on the oligomer cyclic amide 1 ($x=1$; $y=1$), the FCM Panel identified a NOAEL of 1040 mg/kg bw per day, the highest dose tested.

Based on an in silico model that estimated a low $\text{Log } k_{\text{o/w}}$ (1.76) for the cyclic amide 1 and on the intrinsic characteristics (high solubility in water), the accumulation in the human body of the oligomers containing the 2,2'-oxydiethylamine as well as 2,2'-oxydiethylamine itself is very unlikely. In a repeated 28-day oral toxicity study with the cyclic amide 1, levels of the oligomer in the circulating blood remained constant over time, giving a clear indication that a plateau was reached already after 7 days. Moreover, no adverse effects were observed at highest dose tested in the available 90-day toxicity study. Based on the available data, and according to the 'EFSA Note for Guidance', the Panel considered that the potential for accumulation in humans of the oligomers containing the substance is not of concern.

The FCM Panel considered that for a film of maximum 25 μm thickness, the migration at temperatures above 200°C would not be higher than the submitted migration results obtained under conditions representing an extraction rather than a migration (10% ethanol, 6 h at 100°C, followed by 10 days at 60°C).

To cover the risk related to the exposure to oligomers and impurities, the Panel considered it necessary to restrict the uses of the substance to the requested uses, based on the data submitted and assessed.

4 | CONCLUSIONS

The FCM Panel concluded that the substance 2,2'-oxydiethylamine is not of safety concern for the consumer if:

- it has a minimum purity of 99.6% w/w; and
- it is used as a comonomer at up to 14% w/w with adipic acid and caprolactam or their homologues (i.e. with longer C chain) to manufacture polyamide films with a thickness of up to 25 µm; and
- the final films are in contact with any types of foods, except infant formula and human milk, at any time and temperature conditions; and
- the migration of the substance does not exceed 0.05 mg/kg food; and
- the migration of the oligomers below 1000 Da containing the substance does not exceed 5 mg/kg food.

5 | RECOMMENDATIONS

Based on the available migration data and considering that the substance was found to be unstable in 10% ethanol (simulant A) under the applied migration test conditions, the FCM Panel suggested that the compliance with the migration limit is tested using water as worst-case simulant.

6 | DOCUMENTATION AS PROVIDED TO EFSA

Dossier 2,2'-oxydiethylamine. February 2024. Submitted by BASF SE.

ABBREVIATIONS

bw	body weight
CAS	Chemical Abstracts Service
CEF	Panel EFSA Panel on Food Contact Materials, Enzymes, Flavourings and Processing Aids
FCM	Food contact materials
GLP	Good laboratory practice
MoE	Margin of exposure
MoS	Margin of safety
NOAEL	No observed adverse effect level
OECD	Organisation for Economic Co-operation and Development
SCF	Scientific Committee on Food
TTC	Threshold of toxicological concern

AMENDMENT

An editorial correction to the name of the substance on page 4 (first line) was carried out that does not materially affect the contents or outcome of this scientific output. To avoid confusion, the original version of the output has been removed from the EFSA Journal, but is available on request.

An editorial correction related to water as simulant on Chapter 5 - Recommendations (last line) was carried out that does not materially affect the contents or outcome of this scientific output. To avoid confusion, the original version of the output has been removed from the EFSA Journal, but is available on request.

REQUESTOR

German competent authority (Bundesamt für Verbraucherschutz und Lebensmittelsicherheit, BVL)

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